

Project 8 Code Intelligence Assistant for Developers

Domain: Developer Tools / LLM — *Dataset:* github.com/google-research/CodeSearchNet + *kaggle:* [github-code](https://www.kaggle.com/codeparrot/github-code)

Problem statement: Build a VS Code extension and web IDE companion that understands a developer's codebase, suggests contextually relevant code completions, explains unfamiliar functions in plain language, detects potential bugs before compilation, and automatically generates unit tests. This directly competes with GitHub Copilot and replicates the architecture used by Amazon CodeWhisperer.

Dataset: CodeSearchNet Benchmark (github.com/google-research/CodeSearchNet), 6 million code-comment pairs across Python, Java, JavaScript, PHP, Ruby, Go; GitHub Code (*kaggle:* [codeparrot/github-code](https://www.kaggle.com/codeparrot/github-code)).

Placement target: ML Engineer at dev tools companies (JetBrains, GitHub, GitLab, Sourcegraph), software intelligence roles, AI-assisted development teams at all major tech companies.

Student	Sub-problem	Technique	App component
A	Build a code search system using BM25 + dense retrieval (CodeBERT embeddings) on CodeSearchNet. Given a natural language query, retrieve the top-5 relevant code snippets. Evaluate with MRR@5 and NDCG@10.	Code retrieval, CodeBERT embeddings	Code search API
B	Train an AST-based (Abstract Syntax Tree) bug detector: parse Python code into ASTs using <code>ast</code> module, featurise tree structure, train XGBoost to classify buggy vs clean functions on GitHub bug-fix commit data.	AST feature engineering, supervised	Bug detector API
C	Fine-tune CodeT5+ (125M parameter model, fits on free-tier GPU) for code summarisation and unit test generation. Evaluate BLEU-4 and CodeBLEU on the CodeSearchNet test set.	LM fine-tuning, CodeT5+	Code explainer + test generator service
D	Build the VS Code extension (TypeScript) calling a FastAPI backend: inline code explanation on hover, test generation command, bug warning annotations. Deploy backend on Railway with GitHub Actions CI/CD.	VS Code extension, REST API, CI/CD	VS Code extension, web IDE fallback